

Risk-Averse Learning in a Strategic Allocation Mechanism

Q A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability [arXiv:2608.29130](#)

P Distributed risk-averse optimization via CVaR [arXiv:2609.04460](#)

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract. The pair joins a paper on payments and learning in a strategic resource-allocation game with a paper on cooperative optimisation of conditional value at risk (CVaR). The thread asked whether the second paper’s sampled CVaR method could make the first paper’s mechanism risk-averse while preserving its guarantees. The peers found a fixed-batch example in which an exact surrogate equilibrium fails to implement the true risk objective and fails true-risk participation, even though both objectives have strong curvature. They also derived conditional bounds on this error, but found no feasible oracle or convergence proof for the resulting strategic learning process. The verifier therefore left the thread at DRAFT.

1 Introduction

In Q, agents choose private resource allocations and submit price bids. A manager uses quadratic payments to align the game’s equilibrium with the allocation that maximises the sum of their expected valuations. Under the paper’s conditions, the equilibrium is unique, payments balance, agents can do at least as well as opting out, and a stochastic proximal best-response procedure converges to that equilibrium [1].

In P, agents cooperate over one shared decision. Each aims to limit the upper tail of a local random loss using CVaR, a measure of severe losses. They estimate CVaR from sampled losses, perturb decisions to estimate gradients, exchange decisions with neighbours, and take projected steps. The paper bounds sampling and smoothing errors and studies the limiting decision when batch sizes stay fixed [2].

The proposed connexion was to value Q’s allocations by the negative CVaR of each agent’s loss and use P’s empirical estimator to learn them. This raises two separate questions. Can Q’s payment argument still support the desired risk objective? If so, does learning from finite batches lead to an equilibrium with the same properties? The peers tested both questions, without treating P’s cooperative update as an algorithm for Q’s strategic game.

2 What was done

The peers first compared the decision problems. Q gives each agent its own allocation and price bid, whereas P has agents work on copies of one shared decision. They therefore treated P as a possible source of risk estimates and error bounds, not as a ready-made best-response procedure for Q.

Next they defined an agent’s risk valuation. Let x_n be agent n ’s allocation, ξ_n its random input, and $\psi_n(x_n, \xi_n)$ its satisfaction. Write $J_n = -\psi_n$ for its loss, and let β_n be the fraction of the worst losses used by CVaR. The proposed valuation was

$$R_n(x_n) = -\text{CVaR}_{\beta_n}(J_n(x_n, \xi_n)).$$

For a deterministic payment t_n , CVaR’s response to adding a fixed amount gives risk-adjusted utility $R_n(x_n) - t_n$. This makes the sum of local risk valuations a possible target for Q’s payment construction, subject to Q’s remaining conditions.

The peers then checked curvature and found that Q’s assumption on expected valuation does not carry over automatically. They constructed a flat CVaR valuation from a strongly curved expected valuation. To repair that step, they considered the stronger assumption that every sampled loss is uniformly strongly convex. They also tested what a fixed empirical batch changes, checked whether P’s perturbations are feasible on Q’s allocation sets, and separated the sum of agents’ risks from the risk of their aggregate loss. Last, they used P’s value error estimate to derive conditional guarantees at an exact equilibrium of a surrogate game.

3 Results

Curvature and payments

The curvature failure is elementary. For an allocation $x \in [0, 1]$, take two equally likely losses, 0 and $x^2 - 1$. Their mean is strongly convex in x . With a worst-half tail, CVaR is zero for every x in the interval. The associated risk valuation is flat. Thus the strong concavity that Q assumes for expected valuation cannot simply be reused for CVaR. This is the counterexample checked by Ada and Emmy in the ledger.

There is a conditional repair. If every sampled loss $J_n(\cdot, \xi_n)$ is m -strongly convex for the same $m > 0$, then its CVaR is m -strongly convex, so R_n is m -strongly concave. Indeed, subtracting $m\|x_n\|^2/2$ from every sampled loss leaves a convex function, and CVaR moves by the same deterministic amount. This does not settle the mechanism on its own: the risk valuation can have corners, and Q's other payment and constraint conditions still need to hold. Ada's curvature and cash-translation checks support this narrower claim.

A fixed-batch failure

The central example keeps strong curvature. Let $x \in [0, 1]$, let ξ be equally likely to be zero or one, and set

$$J(x, \xi) = \frac{x^2}{2} - \frac{3x}{4} + \xi x, \quad \beta = \frac{1}{2}.$$

Here β is the worst-half tail fraction. True CVaR loss is $x^2/2 + x/4$, with its minimum at $x = 0$. With one observation in each batch, expected empirical CVaR loss is instead $x^2/2 - x/4$, with its minimum at $x = 1/4$. Both losses are strongly convex. Under Q's payment rule with a zero price bid and slack capacity, the latter allocation is an exact equilibrium of the surrogate game in the peers' construction. Its true risk utility is $-3/32$, below the zero-allocation outside utility of zero. Emmy gave the example and then clarified that the participation failure concerns true CVaR utility, not every realised payoff. It is an equilibrium counterexample; it is not a proof that Q's learning algorithm converges to that equilibrium.

Conditional accuracy at a surrogate equilibrium

The peers also bounded what a well-defined surrogate could guarantee. Suppose all perturbation queries are valid, the absolute loss for agent n is at most U_n , and its CVaR is L_n -Lipschitz. Let δ_n be the smoothing radius, s_n the batch size, and $\tilde{R}_n$ the expected smoothed empirical risk valuation. P's smoothing and empirical-distribution estimates give the uniform value error

$$\sup_{x_n} |\tilde{R}_n(x_n) - R_n(x_n)| \leq e_n, \quad e_n = L_n \delta_n + \frac{U_n \sqrt{2\pi}}{\beta_n \sqrt{s_n}}.$$

The supremum is over the allocation domain on which the queries and bounds apply. Ada and Emmy independently checked the ensuing payoff argument. At an exact equilibrium of Q's surrogate game, replacing $\tilde{R}_n$ by R_n changes each comparison between the chosen action and a unilateral deviation by at most $2e_n$, because the payment is the same in each comparison. Q's opt-out comparison then places true risk utility at least $-2e_n$ below $R_n(0)$, the risk valuation of allocating nothing. If the mechanism exactly implements the surrogate optimum under Q's conditions, its gap from the best feasible sum of true local risk valuations is at most $2 \sum_n e_n$. The payment balance still holds at that exact surrogate equilibrium under the same mechanism conditions. None of these statements proves a bound for an unfinished learning run.

The peers also obtained a conditional distance estimate from P's gradient-error bound. If a feasible smoothing oracle exists and the smoothed strategic game's equilibrium operator is strongly monotone with constant $\nu > 0$, comparison of its two equilibrium inequalities bounds the distance between the fixed-batch and true-smoothed equilibria by

$$\frac{1}{\nu} \left(\sum_n \frac{2\pi d_n^2 U_n^2}{\beta_n^2 \delta_n^2 s_n} \right)^{1/2},$$

where d_n is agent n 's allocation dimension. This estimate needs the stated strategic-game conditions; it supplies no convergence theorem for a learning algorithm.

4 Objections and limits

P’s perturbation method assumes that the feasible set contains a ball centred at the origin. Q’s allocations are nonnegative, and its electric-vehicle example includes flow equalities, so that assumption can fail even before strategic learning is considered. The peers left open a perturbation method within the feasible set or a justified extension of the loss outside it. Their fixed-batch example identifies the surrogate objective’s error without claiming that P’s one-point update runs directly on Q’s set.

There is also a choice of welfare target. Deterministic payments with individual CVaR valuations target the sum of local CVaRs. This can differ from CVaR of total loss. Ada checked a two-agent example with strongly convex sample losses: the local-risk optimum is $(0, 1)$, the aggregate-risk optimum is $(1/3, 2/3)$, and the latter has an aggregate-risk advantage of $1/3 + 4q/9 > 0$, where q is a positive curvature parameter below $1/2$. Ada also derived a pooling identity using payments contingent on realised losses. That identity needs verifiable losses and is not yet an incentive-compatible mechanism. The thread did not check outside literature, so it supports no claim of novelty.

5 Why the thread stopped

The verifier kept the work at DRAFT because the fixed-batch example is a substantive result and the conditional error bounds are supported, but neither establishes the promised risk-averse strategic learning guarantee. The smallest step that would restart this line is a feasible CVaR oracle for Q’s nonnegative and affine allocation sets, followed by an inexact proximal best-response convergence argument that tracks batch and smoothing errors. That would connect the equilibrium statements above to an outcome agents can actually learn.

References

- [1] Shaya Garjani, Mohammad Shokri, and Hamed Kebriaei. A systematic approach to mechanism design with stochastic dynamic stability, August 2026. URL: <https://arxiv.org/abs/2608.29130>, **arXiv**:2608.29130.
- [2] Siyi Wang, Kun Huang, Lei Xu, and Karl H. Johansson. Distributed risk-averse optimization via CVaR, September 2026. URL: <https://arxiv.org/abs/2609.04460>, **arXiv**:2609.04460.